Technical report — Gegenprobe: one schema and a converter for Swiss parliamentary answers

Track 2A — OpenParlData · Online · Team 444 — Roman (+ Claude Code agents, §7) · Evidence site https://gegenprobe.mi11ion.io · `make run` re-checks the results (§7) · **Maturity:** census and repair lists = *ready for OPD to apply* (needs its choice on the document hash); nightly job = *deployable prototype* (one real night on a CI runner; not yet at OPD); rules-only converter = *tested prototype*; review mode, whole and virtual affairs = *feasibility study*; schema v0.1 = *draft*.

1. Summary

Verdict. A common schema is feasible with date roles and typed nulls; the converter fills what documents print: the answer date in depth, 7 fields on whole affairs, only the type ready to run unattended (the *unattended bar*: a 95% lower bound ≥ 0.80). Of 17 fields, 12 are shared (meaning tested for 1, the other 11 by definition), 5 shared with a role split, all measured ("answer date" is decided, dispatched or registered), and 3 concepts body-specific; **4 fields are feasible from metadata, 13 partial, 0 need a different concept** (§5). Every value carries its source: page and character span for rule and Apertus dates (an Apertus value is a printed date by construction, located by code), the OPD record for metadata (`route: metadata`); categorical suggestions are document-level, without a span; a missing value is a typed null that says why. By default frozen DE/FR/IT rules run without a model. Outside services: none by default beyond OPD's exports and the parliaments' own sites (fetched politely); review mode adds one model endpoint the operator sets. **The cross-parliamentary result the schema enables.** OpenParlData (OPD) publishes the affairs of 89 Swiss parliaments with affairs in scope (it scrapes 90). Of 91,268 interventions whose events record a written government answer, **16,749 (18.4%) have no answer text in OPD**. 1,422 of them were answered orally in the sitting (1,016 Valais oral-answer stubs, 406 question-time answers), so the other **15,327 are written answers missing: 17.1% of the 89,846 answered in writing** (2024–25: 1,488 of 12,632, 11.8%), each typed by cause. Bern 2024–25: 567 of 593 written answers lack text, most of them in OPD but unlinked; Neuchâtel 2024–25: 161 of 162 are not in OPD. OPD confirmed the Bern linking bug and that our relink list is the workaround until its fix ships (correspondence, 10 Oct 2026; a published excerpt awaits OPD's consent). Deterministic routes propose repairs for 3,107 of the 16,749; 10,800 written answers need per-parliament source adapters. **Where Apertus earns its place** (§6). On 62 documents from 15 parliaments no rule was written for (the 29 answer dates come from 7 of them), Apertus finds 24/29 answer dates, the rules 2/29; per parliament it is ahead in 6 and behind in none (sign test p = 0.031). The latest printed date also finds 22/29 but dates 31 of the 33 answerless documents; Apertus dates 4: its value is finding the date and abstaining. On whole affairs it fills the answer date (22/27 → 27/27; held-out replication 20/35 → 31/35, where the unattended default wrote 4/35 wrong dates, 3 of them by a Luzern rule now guarded in the product path). The local 8B finds the dates too (exploratory, 52 documents without Vaud, fast scorer; it ties the latest printed date, 15 vs 14 of 21) but dates many more answerless documents (§3). Another vendor's model relabelled the Claude-made gold blind: date κ 0.80 and 0.87, no registered decision changes (§6).

2. Architecture

1. **Census** (`census/`): streams the OPD exports once into a per-affair *answer-link status* (`attached | agenda | parentless | absent | image_only | oral`), the feasibility matrix and a comparability measure.
2. **Repair** (`repair/`): deterministic routes per cause, triple-checked (affair number, answer heading, decision date = OPD's answer event), as lists keyed on OPD ids; a nightly job (`deploy/nightly/`) re-runs them.
3. **Converter** (`convert/`): (i) OPD metadata and events, (ii) frozen DE/FR/IT anchor rules, (iii) the Apertus closed-set pointer on the residue. Modes: `rules_only` (default, no model) and `review` (suggestions with page, span and margin, flagged `needs_review`).
4. **Schema and validator** (`schema/`, `validate/`): Pydantic schema v0.1, normative; every field is a value with provenance `{value, role, page, char_start, char_end, quote, route, rule, model_build, θ, margin}` or a *typed null* (`not_in_document | unlinked | image_only | oral | text_failed | present_unanchored | no_rules_for_body | not_applicable`); `QAItem` pairs question *n* with answer *n* (not yet filled); validation uses the Pydantic models (the JSON Schema cannot express cross-field rules).

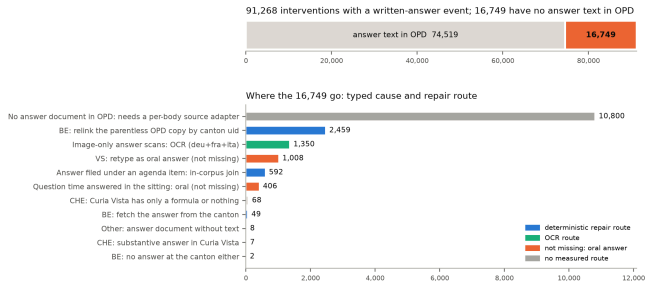


Figure 1. National answer funnel with typed causes and repair routes.

3. Use of Apertus

Closed-set pointer (`convert/pointer.py`, prompts `src/prompts/{de,fr,it}`). The document (first 12,000 characters) and the question go in the user message, the assistant turn is prefilled with a JSON prefix, and `null` and every distinct date printed in the document are scored as continuations by summed log-probability (`prompt_logprobs`). Null if `lp(null) − logsumexp(lp(dates)) > θ`, else the best date, mapped back to its span. θ does not transfer across builds (spike F2: a θ fitted on RLVR scores 0.25 on the final build), so it is fitted per build on dev documents and frozen: θ = 10.40 for the Public AI build (AUROC 0.99, leave-one-out 124/127 [93.3, 99.2]). **Why a pointer** (spikes, `results/report/cited_spikes.json`): asked freely, Apertus dates 15/15 (8B) and 14/15 (70B) Bern interventions that have no answer; a null-allowed prompt fixes the 70B (0/15), not the 8B (14/15).

Which Apertus, which size. Unless marked 8B, Apertus numbers are the **70B** weights, which Public AI serves under the name `swiss-ai/apertus-v1.5-8b`. That is inferred, not documented by Public AI: the served fingerprint (`tp8-pp2-…`, a tensor-parallel 8 × pipeline-parallel 2 deployment) and the behaviour point to the 70B (cited: 760/792 greedy outputs byte-identical to the 70B name; MMLU 84.4% vs 70.8% for the v1 8B route). The `-70b` name answers a system message or prefill with Qwen-SEA-LION (3/3), so forced choice goes to the `-8b` name; the response `model` is checked on every call, and every cached P1/P2 answer comes from one served build (`results/eval/served_models.json`, D10). The **8B** is `swiss-ai/Apertus-v1.5-8B` as a community text-only export (`andreasmartin/apertus-v1.5-8b-text`) quantised to 4 bits by a community member (`Colby/apertus-v1.5-8b-text-Q4_K_M-GGUF`, sha256 `a037df8d…` pinned in `llm.py`; not an official Swiss AI release; `docs/runbook.md` "Supply chain"), run with llama.cpp on one 4-vCPU machine, own θ = 16.73 (pre-registered exploratory arms, D2/D3). It cannot return a date that is not printed (by construction). On P1 it is less accurate than the 70B (pointer and cascade McNemar p = 0.0051 and 3.6×10^{-5}; table in §6) and dates 25 of 90 answerless documents (70B: 7); its residue suggestions are right 7/34 (70B 9/16). On the 52 P2 documents reachable from CI (Vaud's 8 answers missing; D2-6), exploratory: rules 33, pointer 35, cascade 33 of 52 (70B pointer 43, cascade 41; 8B vs 70B 2 vs 10 documents, p = 0.039); it finds 15 of 21 answer dates (70B 16, rules 2, the ungated latest printed date 14) but dates 11 of 31 answerless documents (70B 4). Its fast scorer failed both registered parity gates (57.8, then 4.47 nats) and was admitted through a replacement gate written before any 8B arm ran (bf16 reference, 29/30); its preset self-check failed (10/12) (D9).
Recommendation: `rules_only` unattended; for review suggestions the 8B as the measured sovereign option (a reviewer rejects more), the 70B for accuracy on infrastructure the operator chooses (bf16 weights about 140 GB, e.g. two 80 GB GPUs; arithmetic, not measured). In operation, review mode refuses a served build θ was not fitted on: θ files record the fit's server fingerprints, and the first response from another fingerprint stops the run (exit 3) before any suggestion from it is attached; `docs/runbook.md` "Review mode in operation" gives the refit steps, the owner and the cost.

Cost (`results/report/cost_end_to_end.json`, one core, no GPU). Fetch, extraction and checks take about 2.1 s per document, plus 6.9 s per OCR page; the rules process 480 documents per second (0.5 CPU-hours for all 902,754 documents in OPD's export). Review mode adds the pointer on the residue (97 of 154 held-out documents): 3.5 calls and 5,905 billed prompt tokens per residue document on the 70B, USD 0.59 per 1,000 residue documents at Public AI's list price; the 8B takes 180 s per document on 4 vCPUs (692 CPU-seconds, 9.3 GB RAM), 50 machine-hours per 1,000. On all of OPD's export at that residue share (about 569k documents, an upper bound): about 28,400 8B machine-hours. At ten times OPD's volume: about 5 CPU-hours for the rules, about USD 3,357 for review mode on the 70B.

4. Data and operations

OPD bulk exports (snapshot 6 Oct 2026; OPD's API robots.txt disallows `/v1/`), Bern open data, Fedlex, source PDFs fetched politely (≥ 1.2 s per host; robots.txt per RFC 9309). `make run` and the nightly unit run the image read-only, as a non-root uid, with `--cap-drop ALL` and `no-new-privileges`; PDFs are capped at 50 MB, each

pdftotext/tesseract call at 300 s, each document at 600 s, every hit a typed failure row; failed alerts are re-sent the next night. **One real night** (`results/monitor/night_real_38062530586/`): under systemd on a GitHub-hosted 4-vCPU runner, one network night against OPD's live exports took 83 min (32 of them reading the exports: 6.2 GB of docs/texts streamed, 198 MB of single exports stored; about 50 the first night's triple check of all 2,508 Bern rows), 28 min of CPU, 2,078 MB peak, and reproduced the Bern list on live data (2,459 relink, 49 fetch). It raised no alert; as a first night it had no previous state, so drop and trend alerts could not fire. A two-night run of the recommended Docker setup (night 2 on night 1's state) is scheduled. CI gates both units' `systemd-analyze security` exposure (venv ≤ 2.0, Docker ≤ 5.0; measured 1.2 and 3.9). Inside OPD (`GP_EXPORTS_DIR`) a night with no changed files took 96.8 s on a copied snapshot (0 bytes transferred); one re-scanning three days of changes (80.6% of the docs bytes) about 21.8 min (derived).

5. Results without a model: schema verdict, census, repair

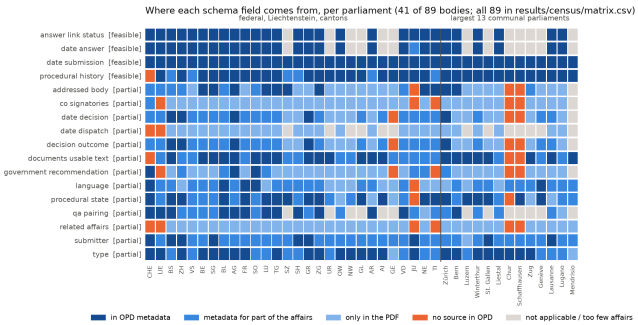


Figure 2. Where each schema field comes from, per parliament (all 89 in `results/census/matrix.csv`). Orange = not in OPD's data for that parliament (the parliament may still publish it), not a different concept.

One verdict per field, on two axes. *Where the value lives*: a field is *feasible* if OPD metadata carries it in ≥ 80% of eligible parliaments, *fundamentally different* (needing a different concept) if ≥ 20% have no written source, else *partial* (thresholds in `docs/census.md` §3, committed with the first matrix); no field is orange in a fifth of the parliaments, at a 10% threshold 5 fields would be (`results/census/tier_sensitivity.json`). *What it means*, from OPD's own vocabularies (`results/census/comparability.json`): the classes below, the finding for the eCH group (`docs/ech-onepager.md`). Type is harmonised for 82.9% of affairs, procedural state for only 278; the government recommendation is in metadata for 34.2% of motions and postulates and PDF-only in 56 bodies.

meaning	feasible from metadata	partial	meaning tested? (evidence)
shared (12)	submission date, link status	submitter, co-signatories, addressed body, recommendation, decision and dispatch dates, Q/A pairing, related affairs, language, usable text	1 of 12: the recommendation (0 of 183 labels ambiguous); the other 11 by definition, untested
shared with a role split (5)	**answer date**, procedural history	type, procedural state, decision outcome	all 5 measured: in 7 of 54 bodies the "answer" event is mostly the submission date; OPD's event equals the printed date for 37 held-out documents and differs for 7; 35 event labels carry different date roles; 42 of 1,038 type labels map to different harmonised types ("Antrag" alone to 7); 32 state, 3 outcome labels ambiguous
body-specific (3)	—	—	oral answers (23.6% of question time is oral); executive-own business; minutes-only councils (16 bodies publish only meeting documents, 8 without any affair → `virtual` affairs, P4)

29.1% of events have no harmonised id. Where OPD metadata carries a field it mostly agrees with print (P3: type 17/18, submitter 23/28, submission date 15/21; affair numbers 21/29, format differences). **Cross-parliamentary question:** across 54 parliaments, 82.3% of written answers have their text in OPD (85.0% after the measured repairs). Answer times are not one quantity across parliaments: pooled over 90,556 dated answers the median is 68 days, but that mixes what OPD's "answer" date is: 112 days where it is the verified decision date (Bern, Ticino: 2,379 of 2,379 answers whose title names the date), 79 where it falls on a government's weekly sitting, 62 for mixed and 39 for diffuse, publication-like dates; in 7 parliaments the "answer" is the submission date.

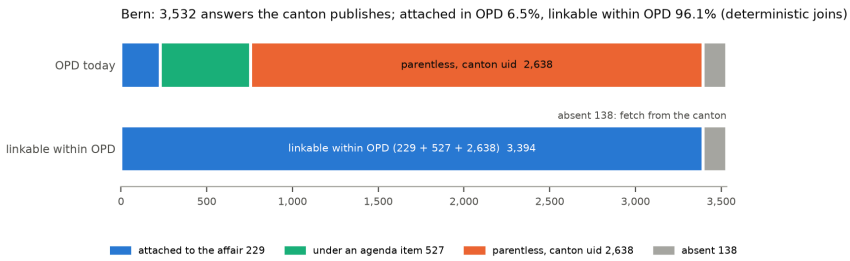


Figure 3. Bern: answers attached to their affair today vs linkable within OPD.

Repair. Bern: of 3,532 answers the canton publishes, OPD attaches 229 (6.5%), although 96.1% are in OPD and linkable: 527 under an agenda item, 2,638 parentless with the canton's document uid; 179 parentless answers are not missing (155 affairs already have the text, 24 have no answer event). Checked against OPD's own copy, all 2,459 relink candidates match affair number and heading; the decision date equals OPD's event for 2,332 with the registered parser and 2,421 [97.9, 98.9] with a parser written afterwards (pre-registered sample: 36/40, three parser misses and one canton typo). **Counts per file:** 2,459 relink candidates = 2,421 in `repair_BE_relink.tsv` + 38 in review; 49 fetch candidates = 48 in `repair_BE_fetch.tsv` + 1 in review (`repair_BE_review.tsv`, 39 rows). `python -m gegenprobe.repair.apply_be` writes a dry-run apply plan (before and after state, guarded statements as text). **Note for OPD:** the document `hash` is sha256(body_key + parent_type + url) (49,810 of 49,810 rows of the 10 Oct export), and the 2,421 relink documents are stored as `parent_type = agenda`, so relinking changes their identifier and file URL unless OPD keeps the old one. A second list corrects 727 Bern `begin_date` load timestamps (557 on 2024-10-28 + 170 from later loads, the groups OPD described; 554 from the affair's own submission event). Federal placeholders (57.3% of federal documents): Fedlex recovers 40/50 [67, 88.8]. Scans: in a probe, DE+FR+IT OCR gave usable text for 15/15 [79.6, 100] image-only answers (a coarse check); re-taking a published text version (Ticino: e.g. `r100.19.doc`) is cheaper.

6. Evaluation: five pre-registrations

Design. Five tests: **P1**, 154 held-out documents from parliaments the rules were written for; **P2**, 62 documents from 15 parliaments no rule was written for; **P3**, 29 whole affairs; **P3b**, a pre-registered replication of P3 on 36 fresh affairs; **P4**, affairs rebuilt from minutes. Every test froze code, prompts, θ and the scorer by sha256 before any label was committed: the LOCK commit precedes the label commit by 6 min 0 s (P1), 2 min 28 s (P2), 3 min 37 s (P3), 4 min 3 s (P4) by commit date (P3b: LOCK5 before the draw), an order GitHub's clock confirms (`results/eval/lock_server_times_all.json`). For P1, the frozen rules were committed together with the held-out sample (D12): P1 is held out by protocol (seeded draw, labels after the lock), and P2 is the clean out-of-sample test of the rules. Unit: one document's answer date, strict. Arms on identical inputs: **rules**; **pointer** (Apertus on every document); **cascade** (rules, then the pointer on the residue = every review suggestion accepted); **naive** free generation, plain and null-allowed, same weights. **Gold, checked by another vendor.** No human labelled a gold value, by design: two blind Claude annotators (Opus, Sonnet), a third breaks ties, quotes verified by code, raw annotations committed. As a check across vendors, `Qwen-SEA-LION-v4-32B-IT` (AI Singapore, Qwen family; neither Anthropic nor Apertus) labelled all 216 P1 and P2 documents blind from the same guide and text (post hoc, D14/D2-7; 216 calls, USD 0.14). Answer present: P1 146/154 (κ 0.89), P2 60/62 (κ 0.93); exact answer date: P1 134/154 (κ 0.80), P2 56/62 (κ 0.87). No registered decision changes when only the documents where both vendors agree are kept or Qwen's labels replace the gold where they differ: P2-H2 is 22 vs 0 under the gold, 21 vs 0 and 23 vs 1. It reads the same extracted text, so it is no human audit (OCR errors stay invisible); disagreements are listed, not adjudicated. P1-H1 (null safety) rests on contested documents: 6 of the 7 values fall on documents OPD types as reports; without the 24 reports it is 1/72 (upper 7.5%), on the documents where both vendors agree 2/84, under Sonnet's labels 5/88. Non-LLM checks: OPD's event matches 37 of 44 checkable P1 dates (5 of the other 7 are 1–5 days apart: decision vs dispatch); Ticino titles 20/20; P3 answer dates 18/21; the 3 P1 gold dates before OPD's submission date are no gold errors (D11); P2: 6 of 29 checkable exactly (6/6 agree), 19/29 face a non-LLM bound. **Statistics** (post hoc, D10): over the 9 registered confirmatory tests all 4 held conclusions survive Holm (P2-H2 3.8×10^{-6}) and BH; the registered lenient metric changes no registered decision; document-level p-values ignore clustering by parliament, so per-parliament results stand beside them. Deviations (`eval/prereg*/DEVIATIONS*.md`) are cited where they apply; D8: the Opus tie-breaker sided with Opus in 36/41 P1 ties, 35 on the recommendation only.

P1 — 154 held-out documents from the parliaments the rules were written for (DE/FR/IT × canton/commune + Ticino scans; 64 with an answer date, 90 without; always-null 90/154; κ 0.95 presence, 0.94 date; 41 tie-breaks).

arm	exact [95% CI]	dates found	wrong printed date	fabricated	values on answerless
rules only (default mode)	**135/154 [81.5, 92]**	47	4	0	2
pointer	130/154 [77.9, 89.3]	46	16	0	6
cascade (suggestions accepted)	139/154 [84.6, 94]	56	11	0	7
8B pointer (CPU, exploratory)	112/154 [65.2, 79.1]	50	37	0	28
8B cascade (CPU, exploratory)	119/154 [70, 83.2]	54	31	0	25
naive, null-allowed	134/154 [80.8, 91.4]	47	11	0	3
↳ null-allowed, re-parsed (D5)	133/154 [80.1, 90.9]	47	13	0	4
naive, plain	98/154 [55.8, 70.8]	47	23	24	39
↳ plain, re-parsed (D5)	103/154 [59.1, 73.8]	54	26	24	41

8B rows (§3; its registered hypotheses: H1 fails; Branch B, FR+IT+scans rules alone right on 10, cascade alone on 2, p = 0.039; H3 AUROC 0.854). 70B: **H1 null safety failed**: the cascade dates 7 of 90 answerless documents (upper bound 15.2% vs 10%). **H2 gave Branch B** (FR+IT+scans, n = 110: cascade alone right on 5 documents, rules alone on 1, p = 0.22): Apertus adds no null safety (cascade 7/90 answerless dated, rules 2/90), so it left the unattended path (default `rules_only`) and stayed as an opt-in review suggestion (D6), which P2 then tested. **H3 passed**: AUROC 0.969 on all documents, 0.966 on the registered length-matched set. **H4 triggered**: 3 of 8 Apertus-decided FR/IT values are wrong. Post hoc: latest printed date on answer-typed documents 131/154 (vs pointer 11:12 documents, p = 1).

P2 — 62 documents from 15 parliaments absent from every dev and held-out set (29 with an answer date, from 7 of the 15: VD 8, NW 6, Bellinzona 5, OW 4, Vernier 3, SZ 2, Schlieren 1; 33 without; always-null 33/62; two Claude annotators κ 1.0 on dates, the other vendor 56/62, κ 0.87). Wilson 95% CIs (`results/eval2/posthoc_r5_p2_table.json`).

arm	exact [95% CI]	dates found [95% CI]	wrong printed date	fabricated	values on answerless [95% CI]
rules only	35/62 [44.1, 68.1]	**2/29** [1.9, 22]	2	0	0 [0, 10.4]
pointer	**53/62** [74.7, 92.2]	**24/29** [65.5, 92.4]	8	0	4 [4.8, 27.3]
cascade	51/62 [71, 89.8]	22/29 [57.9, 87.8]	10	0	4 [4.8, 27.3]
naive, null-allowed	52/62	22/29	7	1	3
↳ null-allowed, re-parsed (D2-4)	54/62	24/29	7	1	3
naive, plain	45/62	21/29	8	4	9
↳ plain, re-parsed (D2-4)	49/62	25/29	9	4	9
latest printed date, ungated (post hoc)	*24/62* [27.6, 51.2]	*22/29* [57.9, 87.8]	*38*	*0*	*31* [80.4, 98.3]
latest date, answer-typed docs (post hoc)	*44/62* [58.7, 80.8]	*18/29* [44, 77.3]	*13*	*0*	*7* [10.7, 37.8]

P2-H1: the rules' recall drops from 47/64 to 2/29. **Per parliament** (post hoc, D2-1, D2-6): the pointer is ahead in 6, the rules in 0, 1 tie on answered documents (sign test p = 0.031); 6 vs 2 with 7 ties on all 62 (p = 0.29). **P2-H2 holds** as registered (document level): on the 29 answered documents, 22 only the pointer vs 0 only the rules (McNemar $p = 4.8 \times 10^{-7}$; without Bellinzona 17 vs 0, $p = 1.5 \times 10^{-5}$); on all 62, 22 vs 4 ($p = 5.3 \times 10^{-4}$). **Baselines** (post hoc): the ungated latest printed date finds 22/29 (pointer vs it on the answered documents 3 vs 1) but dates 31 of the 33 answerless documents; Apertus's value is finding and abstaining. Against the latest date gated by document type (null on member texts): 10 vs 1 documents (p = 0.012), 6 vs 0 parliaments (p = 0.031), a comparison chosen after the results. **P2-H3 failed**: the cascade dates 4/33 answerless documents (upper bound 27.3% vs 15%). **P2-H4 holds**: AUROC 0.983. θ = 0, PREREG2's registered secondary: 54/62. *Error profile:* rules right 35, wrong printed date 2, answerless dated 0, missed 25; pointer 53, 4, 4, 1; cascade 51, 6, 4, 1. The cascade trails the pointer because on 2 documents a rule (`rrb_vom`) fired with a wrong printed date, so Apertus was never asked.

P3 — whole affairs: 29 affairs (58 PDFs), 10 parliaments (5 with rules, 5 without), DE/FR/IT. A feasibility study, not a held-out test: extractor, prompts and sample entered in one commit (D3-5). Arms: OPD metadata (M), rules (R), the default metadata + rules (C), C + Apertus suggestions on fields C leaves open (CA); annotators agreed on ≥ 27/29 affairs per field.

arm	type	number	submitter	submitted	answer date	addressed body*	decision no.†
M	17/29	21/29	23/29	15/21	21/27	0/28	0/17
R	26/29	14/29	5/29	4/21	16/27	4/28	5/17
C	**27/29**	**21/29**	**23/29**	**15/21**	**22/27**	**4/28**	**0/17**
CA	27/29	21/29	23/29	15/21	27/27	23/28	0/17

*Every gold value is "executive"; a constant scores 28/28 (D3-3). †Defect D3-1: the frozen cascade's metadata branch drops the rule's number; post hoc C = 5/17. No field reaches the unattended bar (lower bound ≥ 0.80). H3 (Apertus on open fields: recall 25/32, precision 25/34) is dominated by the near-constant field, withdrawn; without it 6/8 and 6/9. Set fields (C, registered): co-signatories precision 13/16, recall 13/29 (OPD metadata); related affairs 6/9, 6/14 (rules). **P3b, held-out replication** (PREREG5; 36 fresh affairs, 9 parliaments, D5-1; κ ≥ 0.97 per field): *replicated* by its registered criteria (S1: Apertus's recall on the open fields has a lower bound ≥ 0.50; S2: each eligible field within or above P3's CI, at most one below). S2 is lenient: the field it allows below is here the answer date, the one the case for Apertus rests on. Apertus on open fields recall 14/16 [64, 96.5]; 4 of 5 eligible fields within or above P3's CIs; the answer date is the field below (C 20/35; the unattended default wrote 4/35 wrong dates, P1: 4/154; in 3 Luzern affairs a frozen rule replaced OPD's correct date, 23/35 had it been kept, D5-2; with Apertus 31/35, its 11 suggestions 11/11 right); precision 14/20 keeps the reviewer gate; type 35/36 reaches the unattended bar. A product-path guard (not scored) keeps OPD's date when a rule-read date equals the submission date and flags the record: post hoc it fires on 3 of 36 P3b affairs, all Luzern, all kept dates right, none of 29 in P3.

P4 — virtual affairs: 6 held-out sittings of Bulle's council, 120 agenda items. Rules derive 99 items (precision 99/99, titles 99/99), type 91/99 (H2), the decision for 49/63 (46/63 correct), invent none of 36, date all 6 sittings (H4). **H1 failed** (recall 99/120, 82.5%; sub-items, unnumbered lines); **H3 failed** (Apertus +2 decisions, p = 0.5). Transfer cell (2 sittings of La Chaux-de-Fonds, D4-4): items 9/9, titles 9/9, decisions 0/9 by rules, 2/9 found with Apertus (1 correct): decision parsing is commune-specific.

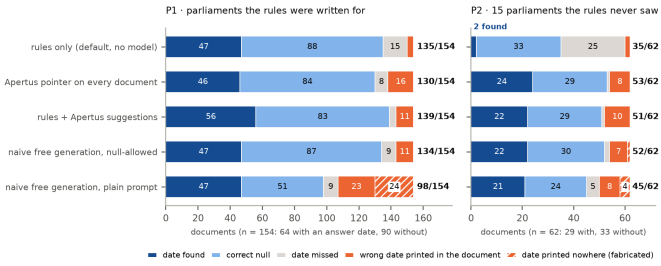


Figure 5. Scoreboards of P1 and P2 (frozen code).

What the tests say. Apertus earns its place where layout varies and no rule exists (P2; the answer date in P3/P3b). On P1 the rules alone match rules + Apertus (H2: Branch B), so the default runs no model, and in P4 Apertus added no significant decisions; the rules are not complete there either (P1 recall 47/64, P4 99/120). **Why the pointer form:** the same Apertus asked freely with a null-allowed prompt is as accurate (P2 re-parsed 54/62 vs 53/62; discordant 1 vs 0, p = 1), but every pointer value is a printed date by construction, its page and span located by code (32/32), and its margin orders the review queue (strong band 26/33 right, weak 3/11, post hoc), while the free prompt quotes its date verbatim for 17/32 values and invents dates printed nowhere (1 in P2; the plain prompt 24 in P1).

7. Limitations, reproducibility, declarations

Answer dates are the field tested in depth; tables, annexes and images are not extracted; P3/P3b are small (29 + 36 affairs), P4 one commune; no gold set contains federal documents; the 8B was not run on P3/P3b; spans index our own pdftotext/tesseract text (`--opd-text` maps them onto OPD's). `make run` (Docker, CPU, clean checkout) recomputes the census, the Bern join and triple check, sample conversions with validator checks, all scores, hypotheses, the 8B analysis and LOCK files, re-derives the P1 rules arm from the source PDFs and P1/P2 Apertus decisions from the model cache, and re-checks every number in `claims.tsv`. Locally: rules 154/154, Apertus 154/154 (P1) and 62/62 (P2) identical; the committed CI record (run 38062904003, `results/ci/`) matched 129/129 and 49/49 reachable decisions and 788 claims, before the newer evidence. Claude Code agents (Anthropic) wrote most code, documentation and the gold labels; numbers are checked by code, not by reading. Public AI (may route outside Switzerland) served evaluation only. Next: the Bern lists and the nightly job at OPD; source adapters for the 10,800 written answers found only at the source; schema v0.2 (date roles, `QAItem`, five gaps) with the eCH group; per-commune decision parsing (P4). Report CC-BY-4.0; sources in §4, standards in `docs/mapping.md`.